CHEN Xiaoyu 陳暁宇

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WORK EXPERIENCE

Specially Appointed Assistant Professor May 2026 – Present
Institute of Science Tokyo Tokyo, Japan
Laboratory for Zero-Carbon Energy (Kato/Takasu Laboratory)

Project Researcher Apr 2025 – Apr 2026
Institute of Science Tokyo Tokyo, Japan

Research focus: Contributing to the Cross-ministerial Strategic Innovation Promotion Program (SIP), 3rd period of SIP "Smart energy management system": developing mobility-based carbon cycle systems focused on CO_2 recovery and resource utilization in e-fuel production and use.

EDUCATION

Nagoya University Apr 2022 – Mar 2025
Ph.D. in Chemical Systems Engineering Nagoya, Japan
Research: Redox thermochemical energy-storage modules and systems; reactor design and numerical simulation.

Nagoya University Apr 2020 – Mar 2022
M.Eng. in Chemical Systems Engineering Nagoya, Japan
Research: Redox thermochemical energy-storage materials, metal oxides, and gas–solid reaction kinetics.

Huizhou University Sep 2015 – Jun 2019
B.Eng. in Chemical Engineering and Technology Huizhou, China
Research: Metal-oxide anode materials for lithium-ion batteries.

RESEARCH & TEACHING EXPERIENCE

JSPS Research Fellowship for Young Scientists (DC2) Apr 2023 – Mar 2025
Nagoya University, Kita/Kubota Laboratory Research Fellow
Designed redox thermochemical storage modules and reactors for 500–1000 °C operation; optimized module geometry and system conditions through numerical simulation.

Aichi Knowledge Hub Priority Research Project III Apr 2021 – Mar 2022
Nagoya University, Kita/Kubota Laboratory Research Assistant
Investigated thermal/electric-battery energy management, including operating-temperature control and screening of medium- to high-temperature storage materials.

Practical Data Scientist Training Program May 2022 – Mar 2025
Nagoya University Mathematical and Data Science Center Qualified Teaching Assistant
Supported coursework and guided student teams in solving industry-provided, real-world data-science problems.

Hokkaido University, Faculty of Engineering Jul 2024 – Sep 2024
Center for Advanced Research of Energy and Materials Visiting Doctoral Student
Applied machine learning to dopant optimization for controlling the operating temperature of redox thermochemical storage materials.

SELECTED PUBLICATIONS (FIRST AUTHOR)

1. **X. Chen**, et al., Heat Release Demonstration of a Novel $CuMn_2O_4/CuMnO_2$-Based Honeycomb Structure Module for Thermochemical Energy Storage. *ACS Sustainable Chem. Eng.*, 13(15), 5580–5591 (2025).
2. **X. Chen**, et al., Development of Redox-type Thermochemical Energy Storage Module: A Support-Free Porous Foam Made of $CuMn_2O_4/CuMnO_2$ Redox Couple. *Chem. Eng. J.*, 485, 149540 (2024).
3. **X. Chen**, et al., An In-depth Oxidation Kinetic Study of $CuCr_xMn_{1-x}O_2$ ($x = 0, 0.1, 0.3$) for Thermochemical Energy Storage at Medium-high Temperature. *Sol. Energy Mater. Sol. Cells*, 260, 112495 (2023).
4. **X. Chen**, et al., Effect of Cr Addition on Cu–Mn Spinel/Delafossite Redox Couples for Medium-High Temperature Thermochemical Energy Storage. *ACS Appl. Energy Mater.*, 5(5), 5811–5821 (2022).
5. **X. Chen**, et al., Exploring Cu-Based Spinel/delafossite Couples for Thermochemical Energy Storage at Medium-high Temperature. *ACS Appl. Energy Mater.*, 4(7), 7242–7249 (2021).
6. **X. Chen**, et al., Investigation of Sr-based Perovskites for Redox-type Thermochemical Energy Storage Media at Medium-high Temperature. *J. Energy Storage*, 38, 102501 (2021).

SELECTED INTERNATIONAL CONFERENCE PRESENTATIONS

1. **X. Chen**, et al., Demonstration of Cu–Mn Composite-Oxide Honeycomb Modules for Medium-High-Temperature Thermochemical Energy Storage. *IMPRES 2025*, Sendai, Oct 2025.
2. **X. Chen**, et al., Development of $CuMn_2O_4/CuMnO_2$-Based Porous-Structure Thermochemical Energy Storage Modules. *3rd ACTS*, Shanghai, Jun 2024.
3. **X. Chen**, et al., Oxidation Kinetics of Cu–Cr–Mn Spinel/Delafossite for Redox Thermochemical Energy Storage at Medium-High Temperature. *Materials Today Conference 2023*, Singapore, Aug 2023.
4. **X. Chen**, et al., Screening and Development of Redox Thermochemical Energy Storage Materials for Medium-High Temperatures. *IMPRES 2022*, Barcelona, Spain, Oct 2022.
5. **X. Chen**, et al., Screening of Redox Chemical Heat Storage Materials for Medium-High Temperatures. *2nd ACTS*, Fukuoka, Japan, Oct 2021.

HONORS & AWARDS

- All-Japan Federation of Overseas Chinese Professionals Scholarship (Aug 2024)
- Nagoya University Outstanding Graduate Student Award (Jul 2024)
- JSPS Research Fellowship for Young Scientists, DC2 (Apr 2023 – Mar 2025)
- Nitto Academic Promotion Foundation, 39th Overseas Dispatch Grant (Nov 2022)
- Nagoya University Akasaki Student Award (Aug 2022)
- Tokai National University Organization, Fusion-Oriented Next-Generation Researcher (Apr 2022 – Mar 2023)
- Nagoya University Chemical Systems Engineering, Master's Thesis Presentation Award (Mar 2022)
- Society of Automotive Engineers of Japan, Graduate Research Award (Mar 2022)